Test Series Code

02

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | **DROPPER** | **M. MARKS: 720**

DATE: 02/11/2025

SYLLABUS

Physics:	Newton's Laws of Motion, Work, Energy & Power, Centre of Mass & System of Particles
Chemistry:	Atomic Structure, Solutions
Botany:	Cell Cycle & Cell Division
Zoology:	Breathing and Exchange of Gases, Body Fluids and Circulation

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. Two blocks of masses 4 kg and 6 kg are connected at the two ends of a light inextensible string passing over a frictionless pulley. The accelerations of 4 kg and 6 kg blocks are $\vec{a}_1$ and $\vec{a}_2$ respectively. Then the value of $|\vec{a}_1 - \vec{a}_2|$ is: ($g = 10\ m\ s^{-2}$)

(1) zero
(2) $2\ m\ s^{-2}$
(3) $4\ m\ s^{-2}$
(4) $8\ m\ s^{-2}$

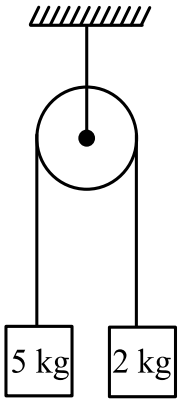
2. A man of mass 50 kg stands on a weighing machine in a lift. The position y of lift (in m) as function of time t (in s), is given by $y(t) = 5 + 5t$. The reading of the weighing machine is:

(1) 25 kg-wt
(2) 75 kg-wt
(3) 50 kg-wt
(4) 100 kg-wt

3. A car starts from rest to cover a distance s on a horizontal road. The coefficient of friction between the tyre and the road is μ. The minimum time in which the car can cover the distance s is proportional to:
(assume car use maximum friction limited acceleration)

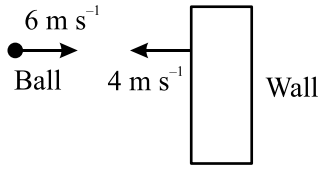
(1) $\sqrt{\mu}$ (2) μ
(3) $\frac{1}{\mu}$ (4) $\frac{1}{\sqrt{\mu}}$

4. Two blocks as shown in the figure are held at rest. String is massless and inextensible, and pulley is frictionless. The magnitude of acceleration of centre of mass of the system, when blocks are released, is: ($g = 10\ m\ s^{-2}$)



(1) $\frac{70}{9}\ m\ s^{-2}$ (2) $\frac{90}{49}\ m\ s^{-2}$
(3) $\frac{10}{7}\ m\ s^{-2}$ (4) $\frac{30}{7}\ m\ s^{-2}$

5. A ball shown in figure collides inelastically ($e = 1/3$) with a heavy moving wall. Speed of the ball just after the collision is:

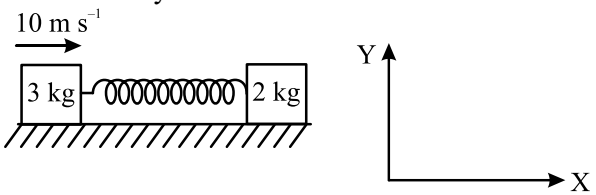


(1) $10\ m\ s^{-1}$ (2) $\frac{10}{3}\ m\ s^{-1}$
(3) $\frac{22}{3}\ m\ s^{-1}$ (4) $2\ m\ s^{-1}$

6. Work done by a body is represented by the equation $W = (2t^2 + 3t + 4)$ J (where t is time in second) then the instantaneous power of the body at $t = 2$ second will be;

(1) 15 W (2) 13 W
(3) 12 W (4) 11 W

7. Two blocks of masses $m_1 = 2$ kg and $m_2 = 3$ kg are attached with an ideal spring of force constant $k = 10$ N/m. The whole system is placed on a smooth horizontal surface. An impulsive force on 3 kg block provides a velocity of $10\ m\ s^{-1}$ towards 2 kg block, as shown in the figure. The velocity of the centre of mass of the system is:



(1) $6\hat{i}\ m\ s^{-1}$ (2) $-6\hat{i}\ m\ s^{-1}$
(3) $10\hat{i}\ m\ s^{-1}$ (4) $-10\hat{i}\ m\ s^{-1}$

8. The potential energy function associated with a conservative field is given by.
$U = -2x^2y$ + constant
Then conservative force is given by:

(1) $\vec{F} = 4xy\hat{i} + x^2\hat{j}$
(2) $\vec{F} = 4xy\hat{i} + 2x^2\hat{j}$
(3) $\vec{F} = 2xy\hat{i} + 2x^2\hat{j}$
(4) $\vec{F} = 2xy\hat{i} + x^2\hat{j}$

9. A moving mass of 8 kg collides head-on perfectly elastically with a stationary mass of 2 kg. If K_0 be the kinetic energy of the moving mass 8 kg before collision, then kinetic energy left with 8 kg mass after the collision will be:

(1) $0.64\ K_0$ (2) $0.80\ K_0$
(3) $0.08\ K_0$ (4) $0.36\ K_0$

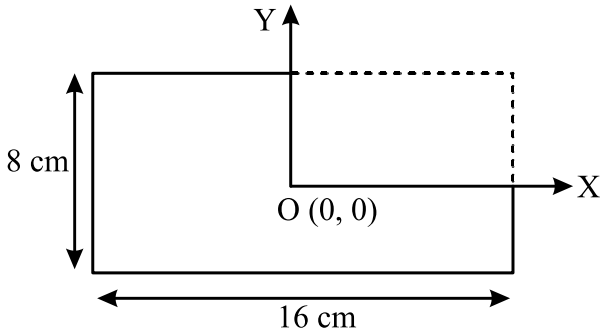
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10. A stone of mass 1 kg is tied to the end of a massless string 1 m long. It is whirled in a vertical circle. If the speed of the stone at the topmost point be 4 m s^{-1}, find the tension in the string at the lowest point. ($g = 10$ m s^{-2})

(1) 6 N (2) 76 N

(3) 52 N (4) 66 N

11. From a 16 cm × 8 cm rectangular uniform plane sheet, exactly one-fourth of the sheet is removed as shown in the figure. If the origin be taken at the center of the original sheet, the co-ordinates of the center of mass of the remaining sheet is:



(1) $\left(-\frac{4}{3},-\frac{2}{3}\right)$ cm

(2) $\left(-\frac{8}{3},-\frac{2}{3}\right)$ cm

(3) $\left(-\frac{4}{3},-\frac{4}{3}\right)$ cm

(4) $\left(-\frac{8}{3},-\frac{8}{3}\right)$ cm

12. A man is standing on a rough ($\mu = 0.5$) horizontal disc rotating with constant angular velocity 5 rad s^{-1}. At what distance r from the centre of disc, should he stand so that he does not slip on the disc? ($g = 10$ m s^{-2})

(1) $r \leq 0.5$ m

(2) $r \leq 0.2$ m

(3) $r \leq 1$ m

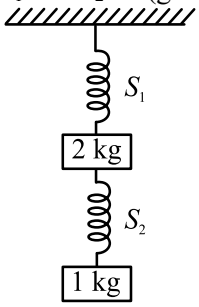
(4) $r > 0.2$ m

13. A block slides down a rough inclined plane of slope angle 37° with constant velocity. It is then projected up from the bottom of same plane with an initial velocity 20 m s^{-1}. Upward distance travelled by block before it comes to rest, will be ($g = 10$ m s^{-2})

(1) $\frac{50}{3}$ m (2) $\frac{200}{3}$ m

(3) $\frac{200}{9}$ m (4) $\frac{100}{3}$ m

14. In the figure shown, two springs S_1 and S_2 are ideal. when only spring S_1 is cut, then immediate acceleration of 2 kg block is a_1 but when only spring S_2 is cut, then its acceleration is a_2. The ratio of magnitude of a_1 and a_2 is: ($g = 10$ m s^{-2})



(1) $\frac{1}{3}$ (2) 2

(3) $\frac{1}{2}$ (4) 3

15. A rocket with a lift-off mass 25,000 kg is blasted upwards with an initial acceleration of 6 m s^{-2}. The initial thrust of the blast is: ($g = 10$ m s^{-2})

(1) 4×10^4 N (2) 4×10^5 N

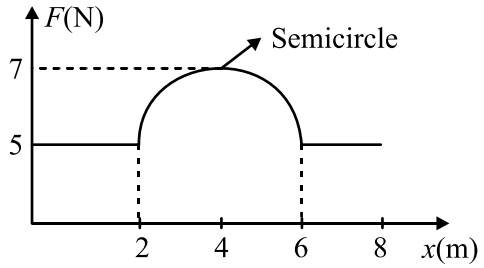
(3) 4×10^6 N (4) 1.5×10^5 N

16. A body of mass 3 kg is acted upon by a force. The displacement s (in m) is given by $s = \frac{1}{3}t^3$, where t is time in second. Work done by the force in first 2 seconds is:

(1) 24 J (2) 12 J

(3) 36 J (4) 48 J

17. The variation of force F with position x of particle is shown in F-x graph. The work done by the force in displacing a body from $x = 2$ m to $x = 6$ m is ($\pi = 3.14$)



(1) 6.28 J

(2) 24.74 J

(3) 28.24 J

(4) 26.28 J

18. A body is initially at rest. It undergoes one dimensional motion with constant force. The instantaneous power delivered to it at time t is proportional to:

(1) $t^{1/2}$ (2) t

(3) $t^{3/2}$ (4) t^2

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19. The blades of a windmill sweep out a circle of area $A = 30\ m^2$. The wind flows with velocity $v = 36\ km\ h^{-1}$ perpendicular to the circle. Assuming that the windmill converts 25% of wind's energy into electrical energy, what is the electrical power produced? (Given density of air, $\rho = 1.2\ kg\ m^{-3}$)

(1) 9 kW (2) 6 kW
(3) 4.5 kW (4) 12 kW

20. Given below are two statements:

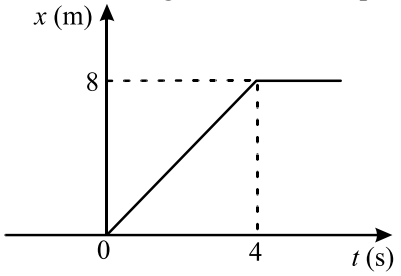
Statement I: A body of mass 1 kg is making 1 rps in a circle of radius 2 m. Centripetal force acting on it is $4\pi^2$ N.

Statement II: Centripetal force on a body moving in a circle is given by $F = m\omega^2 r$. (symbols have usual meaning)

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

21. Figure shows the position-time ($x - t$) graph of a particle of mass 2 kg. What is the impulse at $t = 4$ s?



(1) 4 N s (2) –4 N s
(3) 8 N s (4) –8 N s

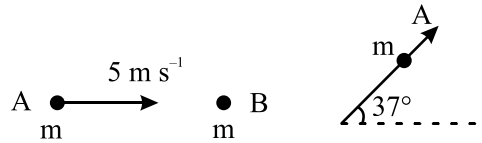
22. A block of mass m = 1 kg is kept on a rough inclined plane of inclination angle $\theta = 30°$. Coefficient of static friction between the block and the plane is 0.8. Match the entries of **List-I** with correct entries in **List-II**. ($g = 10\ ms^{-2}$)

List-I		List-II	
A.	Contact force between the block and the plane	I.	$4\sqrt{3}$ N
B.	Actual frictional force acting between the block and the plane	II.	1.93 N
C.	Limiting force of friction between the block and the plane	III.	10 N
D.	Minimum force required along the plane to just move the block down the plane	IV.	5 N

Choose the **correct** answer from the options given below.

(1) A-III; B-IV; C-II; D-I
(2) A-I; B-IV; C-II; D-III
(3) A-I; B-II; C-IV; D-III
(4) A-III; B-IV; C-I; D-II

23. A particle *A* of mass *m* moving with velocity 5 m s^{-1} along *x*-axis. It makes oblique perfectly elastic collision with another identical mass *B* at rest. After the collision, particle *A* deflects at an angle 37° as shown. Speed of mass *B* immediately after the collision is:



(1) 5 m s^{-1} (2) 4 m s^{-1}
(3) 3 m s^{-1} (4) $2\sqrt{2}$ m s^{-1}

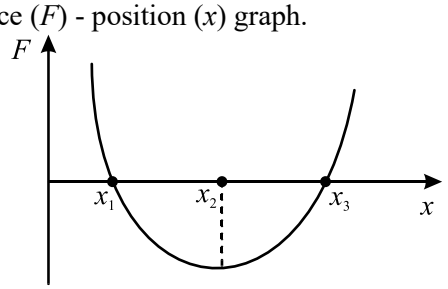
24. Which of the followings is **not** a conservative force?

(1) Gravitational force
(2) Spring force
(3) Electrostatic force
(4) Viscous force

25. The upper $\left(\frac{2}{3}\right)^{rd}$ part of a fixed inclined plane of inclination θ is perfectly smooth while lower $\left(\frac{1}{3}\right)^{rd}$ part is rough. A block starting from rest at the top of the plane, comes to rest just at the bottom. The coefficient of kinetic friction between the block and the lower $\left(\frac{1}{3}\right)^{rd}$ part of the plane is:

(1) $\mu = 2\tan\theta$ (2) $\mu = \frac{\tan\theta}{2}$
(3) $\mu = \frac{2\tan\theta}{3}$ (4) $\mu = 3\tan\theta$

26. Consider the statements followed by conservative force (F) - position (x) graph.



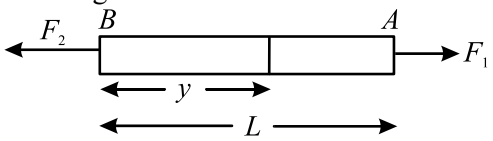
(A) At $x = x_1$, particle is in stable equilibrium
(B) At $x = x_2$, particle is in stable equilibrium
(C) At $x = x_3$, particle is in unstable equilibrium

Correct statement(s) is/are:

(1) A, B and C (2) B and C only
(3) A and C only (4) A and B only

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27. A uniform rod of mass M and length L is acted on by two unequal forces F_1 and F_2 ($F_1 > F_2$) as shown in the figure.



The tension in the rod at a distance y from the end B is given by

(1) $F_1\left(\frac{y}{L}\right)+F_2\left(1-\frac{y}{L}\right)$

(2) $F_1\left(1-\frac{y}{L}\right)+F_2\left(\frac{y}{L}\right)$

(3) $(F_1-F_2)\left(\frac{y}{L}\right)$

(4) $(F_1-F_2)\left(\frac{L}{y}\right)$

28. A cricket ball of mass 1 kg has an initial velocity $\vec{u}=(4\hat{i}+3\hat{j})$ m s^{-1} and a final velocity $\vec{v}=-(4\hat{i}+3\hat{j})$ m s^{-1} after being hit. The magnitude of change of momentum of the ball is;

(1) 10 N s (2) 14 N s
(3) 5 N s (4) zero

29. A uniform chain of mass m and length l is held on a frictionless table with one-third of its length hanging over the edge. The work done in just pulling the hanging part back on the table slowly is: (g is acceleration due to gravity)

(1) $\frac{mgl}{72}$ (2) $\frac{mgl}{36}$
(3) $\frac{mgl}{3}$ (4) $\frac{mgl}{18}$

30. A cracker is thrown in air with speed 20 m s^{-1} at angle 45° with the horizontal. At a height 1 m from the ground, it explodes into a number of fragments which follow different parabolic paths. What is the speed of centre of mass of all fragments, when it is at a height 2 m from the ground? (g = 10 m s^{-2})

(1) $6\sqrt{10}$ m s^{-1} (2) $6\sqrt{5}$ m s^{-1}
(3) $4\sqrt{10}$ m s^{-1} (4) $6\sqrt{3}$ m s^{-1}

31. The dimensional formula of Power is:

(1) [ML^{2}T^{-2}] (2) [ML^{-2}T^{-3}]
(3) [ML^{2}T^{-3}] (4) [MLT^{-3}]

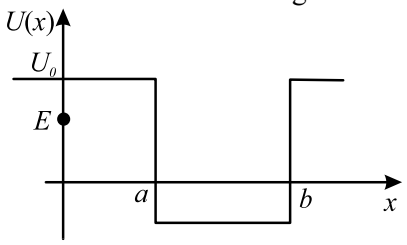
32. A ball is dropped from a height h. If the coefficient of restitution is e = 0.5 and height attained by the ball after the collision is 1 m, then the value of h is:

(1) 2 m (2) 0.25 m
(3) 4 m (4) 8 m

33. A cyclist skids (both wheels locked) for 5 m before coming to a stop. During this process, the force on the cycle due to the road is 200 N and is directly opposed to the motion. The work done by the cycle on the road is:

(1) 1000 J (2) –1000 J
(3) –500 J (4) zero

34. Figure shows potential energy $U(x)$ varies with position x. The total energy E is indicated by a dot on the y axis. Specify in which region the particle **cannot** be found for the given total energy E?



(1) $x < a, x > b$ (2) $a < x < b$
(3) $x > \frac{a}{2}$ (4) $x > 2b$

35. A car is running on a circular track of radius R and banking angle θ. If the coefficient of friction between the tyres and the road is μ_s ($\tan\theta > \mu_s$), then the minimum speed to avoid slipping is: (g is acceleration due to gravity)

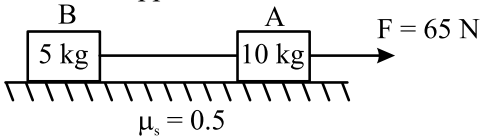
(1) $\left[\frac{Rg(\mu_s+\tan\theta)}{(1-\mu_s\tan\theta)}\right]^{1/2}$

(2) $\left[\frac{Rg(\mu_s+\tan\theta)}{(1+\mu_s\tan\theta)}\right]^{1/2}$

(3) $\left[\frac{Rg(\tan\theta-\mu_s)}{(1-\mu_s\tan\theta)}\right]^{1/2}$

(4) $\left[\frac{Rg(\tan\theta-\mu_s)}{(1+\mu_s\tan\theta)}\right]^{1/2}$

36. Two blocks A and B of masses 10 kg and 5 kg respectively are connected by a massless string and placed on a rough horizontal surface. The coefficient of static friction between the surface and both blocks is μ_s = 0.5. An external horizontal force of 65 N is applied on the block A.

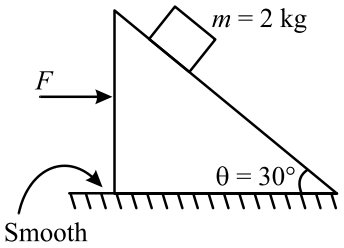


If T is the tension in the string and force of friction between the block B and horizontal surface is f_B. Then values of T and f_B respectively are:
(g = 10 m s^{-2})

(1) 7.5 N, 7.5 N (2) 0 N, 15 N
(3) 15 N, 15 N (4) 15 N, 25 N

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37. Figure shows a block of mass $m = 2$ kg kept over a smooth inclined plane of angle $\theta = 30°$. A horizontal force F is applied on the inclined plane such that block remains stationary w.r.t. the plane.
($g = 10$ m/s^2)

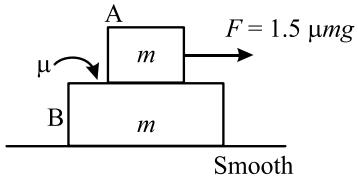


With respect to the non-inertial reference frame of the moving inclined plane, net force acting on the block is:
($g = 10$ m s^{-2})

(1) 0 N (2) $20\sqrt{3}$ N

(3) $10\sqrt{3}$ N (4) $\frac{20}{\sqrt{3}}$ N

38. Two blocks A and B of equal masses m are placed on a smooth horizontal surface as shown in the figure.



The coefficient of static and kinetic friction between the two blocks is μ. A horizontal force $F = 1.5\ \mu mg$ is applied on block A. Acceleration of block A is a_1 and that of block B is a_2. Then:
(g is acceleration due to gravity)

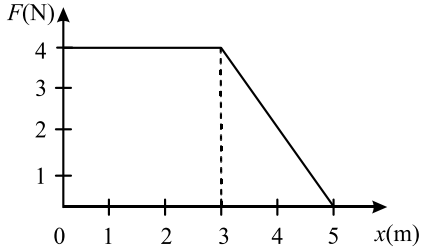
(1) $a_1 = a_2 = \frac{3}{4}\mu g$

(2) $a_1 = \frac{\mu g}{2}, a_2 = \mu g$

(3) $a_1 = \mu g, a_2 = \frac{\mu g}{2}$

(4) $a_1 = \frac{\mu g}{2}, a_2 = 0$

39. A force F (in N) acting on an object varies with distance x (in m) as shown in the graph. The work done by the force in moving the object from $x = 0$ to $x = 2$ m is:



(1) 12 J (2) 8 J

(3) 16 J (4) 20 J

40. A ball of mass $2m$ moving with velocity $v\hat{i}$ makes head-on collision with a ball of mass m moving with velocity $-v\hat{i}$. If collision is perfectly inelastic, then loss of kinetic energy of the system due to collision will be:

(1) $\frac{4}{3}mv^2$ (2) $\frac{2}{3}mv^2$

(3) $\frac{8}{3}mv^2$ (4) $\frac{1}{2}mv^2$

41. Two bodies of masses m and M ($M > m$) are having equal magnitude of momentum. The ratio of their respective kinetic energy $\left(\frac{K_m}{K_M}\right)$ is:

(1) $\frac{m}{M}$ (2) $\frac{M}{m}$

(3) $\frac{m^2}{M^2}$ (4) $\sqrt{\frac{M}{m}}$

42. An elevator can carry a maximum load of 1800 kg (elevator + passengers) is moving up with a constant speed of 1 m s^{-1}. The frictional force opposing the motion is 2000 N. The power delivered by the motor to the elevator is: ($g = 10$ m s^{-2})

(1) 44 kW (2) 20 kW

(3) 22 kW (4) 10 kW

43. In a collision, total linear momentum of the system remains conserved in:

(1) perfectly elastic collision only
(2) perfectly inelastic collision only
(3) inelastic collision only
(4) any type of collision

44. Two balls of masses 2 m and 4 m are thrown vertically upward with speed 20 m s^{-1} and 10 m s^{-1} respectively. The magnitude of acceleration of centre of mass during upward journey of both balls is:
($g = 10$ m s^{-2})

(1) $\frac{40}{3}$ m s^{-2} (2) 10 m s^{-2}

(3) $\frac{20}{3}$ m s^{-2} (4) $\frac{10}{3}$ m s^{-2}

45. When an ideal spring is stretched by x from its natural length, work done is W. When it is further stretched by amount x, additional work done will be:

(1) W (2) $4W$

(3) $6W$ (4) $3W$

[Contd...

46. The smallest wavelength of radiations that can be emitted by H-atom is: (Where, R = Rydberg constant)

(1) $\frac{1}{R}$ (2) $\frac{1}{4R}$

(3) $\frac{1}{9R}$ (4) $\frac{1}{2R}$

47. van′t Hoff factor of a solute (AB) in a solvent where it is 75% dissociates into A^+ and B^- and remaining trimerises, is:

(1) 1.74 (2) 1.92

(3) 1.20 (4) 1.58

48. The ratio of frequencies of light having $\lambda_1 = 4.1\times10^{-5}$ cm to that of light having $\lambda_2 = 2.05\times10^{-5}$ cm is:

(1) 2:1 (2) 1:1

(3) 1:2 (4) 1:1.4

49. Mass of NaCl required to decrease the freezing point of 1860 g water (K_f = 1.86 K kg mol^{-1}) to –1°C is (Assume NaCl dissociates completely):

(1) 25.12 g (2) 37.10 g

(3) 29.25 g (4) 16.16 g

50. A radiation of wavelength λ illuminates a metal and ejects photoelectrons of maximum kinetic energy of 1 eV. Another radiation of wavelength $\frac{\lambda}{2}$, ejects photoelectrons of maximum kinetic energy of 4 eV. The work function of metal is:

(1) 2.5 eV (2) 2 eV

(3) 3 eV (4) 3.5 eV

51. Given below are two statements:

Statement I: Osmosis always takes place from hypertonic to hypotonic solutions.

Statement II: The release of nitrogen gas dissolved in the blood creates a medical condition known as anoxia.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are incorrect.

52. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:

Assertion A: Half filled and completely filled subshells have extra stability.

Reason R: It is due to relatively small shielding, smaller coulombic repulsion energy and larger exchange energy.

In the light of above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

53. The elevation in boiling point of a very dilute solution containing non-electrolyte solute in water at 1 atm pressure is 0.26°C. The vapour pressure of the solution (in mm of Hg) is:

(Given: K_b of H_2O = 0.52 K kg mol^{-1})

(1) 753.16

(2) 758.12

(3) 756.56

(4) 751.24

54. The **correct** set of quantum numbers for the last electron of Be (Z = 4) atom is:

(1) $n=1, \ell=0, m=0, s=+\frac{1}{2}$

(2) $n=2, \ell=1, m=0, s=-\frac{1}{2}$

(3) $n=2, \ell=0, m=+1, s=+\frac{1}{2}$

(4) $n=2, \ell=0, m=0, s=-\frac{1}{2}$

55. Match **List-I** with **List-II**.

List-I (Gas)		List-II (K_H in water at 298 K)	
(A)	Methane	(I)	40.3 kbar
(B)	Vinyl chloride	(II)	1.83×10^{-5} kbar
(C)	Formaldehyde	(III)	0.413 kbar
(D)	Argon	(IV)	0.611 kbar

Choose the **correct** answer from the options given below:

(1) A-III; B-IV; C-II; D-I

(2) A-III; B-I; C-IV; D-II

(3) A-IV; B-I; C-II; D-III

(4) A-IV; B-II; C-I; D-III

[Contd...

56. Which of the following represents increasing order of frequency for given radiations?
(1) Radiowaves, UV-rays, X-rays
(2) X-rays, Radiowaves, UV-rays
(3) UV-rays, X-rays, Radiowaves
(4) UV-rays, Radiowaves, X-rays

57. Amalgam of mercury with sodium is a type of:
(1) solid in solid solution
(2) solid in liquid solution
(3) liquid in liquid solution
(4) liquid in solid solution

58. $X_2(g)$ molecule dissociate into X(g) atoms after absorbing light of 6620 Å. If one quantum of radiation is absorbed by each $X_2(g)$ molecule, then the kinetic energy of X(g) atom will be:
(Given: Bond energy of $X_2(g) = 60.22$ kJ mol^{-1}, h = 6.62×10^{-34} Js)
(1) 1×10^{-19} J
(2) 3.5×10^{-19} J
(3) 2×10^{-19} J
(4) 0.5×10^{-19} J

59. The molality of 20% (mass/mass) aqueous solution of glucose having density 1.2 g mL^{-1} is:
(1) 1.39 m (2) 0.86 m
(3) 2.12 m (4) 2.94 m

60. The **correct** order of energies of given orbitals for He^+ ion is:
(1) 3p < 4s < 3d (2) 3p = 3d = 4s
(3) 3p < 3d = 4s (4) 3p = 3d < 4s

61. 9.8% (w/w) H_2SO_4 stock solution has a density 2 g/mL. The volume of the stock solution required to prepare a 200 ml solution of 0.1 M H_2SO_4 solution is:
(1) 5 ml (2) 25 ml
(3) 10 mL (4) 20 ml

62. The radius of 2nd excited state of Be^{3+} ion is:
(1) 119.03 pm (2) 158.7 pm
(3) 238.05 pm (4) 79.35 pm

63. Osmotic pressure of an aqueous solution of 0.1 M monobasic acid having pH = 1 at temperature T K will be:
(1) 1.1 RT
(2) 0.2 RT
(3) 0.1 RT
(4) RT

64. Which among the following phenomena can explain the particle nature of electromagnetic radiation?
(A) Interference
(B) Black body radiation
(C) Photoelectric effect
(D) Line spectra of atoms with special reference to hydrogen.
(1) A, B and C only (2) B and C only
(3) B, C and D only (4) C only

65. For an ideal binary solution of liquids A and B, the slope of plot between $\left(\frac{1}{p_{total}}\right)$ and Y_A is: (where Y_A is the mole fraction of A in vapour phase)
(1) $\frac{1}{p_A^0}$ (2) $\frac{1}{p_A^0} - \frac{1}{p_B^0}$
(3) $\frac{1}{p_B^0}$ (4) $\frac{1}{p_B^0} - \frac{1}{p_A^0}$

66. Given below are two statements:
Statement I: The amount of light emitted (intensity of radiation) from a black body depends only on its temperature.
Statement II: Number of neutrons present in the nucleus have very strong effect on the chemical properties of an element.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

67. Which of the following is/are **not** correct for an ideal solution?
(A) $\Delta_{mix} H = 0$
(B) $\Delta_{mix} V = 0$
(C) $\Delta_{mix} G = 0$
(D) $\Delta_{mix} S = 0$
(1) A and B (2) C only
(3) B and C (4) C and D

68. The number of spectrum lines obtained in visible region of H-atom spectrum when electrons de-excite from 5th orbit to ground state is:
(1) 5 (2) 3
(3) 10 (4) 4

[Contd...

69. Consider the following statements regarding vapour pressure of aqueous solutions containing non-volatile, non-electrolyte solutes:
(A) It is a colligative property
(B) It increases with increase in temperature
(C) It has same value for isotonic solutions at same temperature
The **correct** statement(s) is/are:
(1) A and B (2) B and C
(3) A and C (4) B only

70. Match **List-I** with **List-II**.

List-I (Atom)		List-II (Unpaired electron)	
(A)	Pd(Z = 46)	(I)	6
(B)	Cr(Z = 24)	(II)	2
(C)	Ag(Z = 47)	(III)	0
(D)	Ni(Z = 28)	(IV)	1

Choose the **correct** answer from the options given below:
(1) A-III; B-II; C-IV; D-I
(2) A-III; B-I; C-IV; D-II
(3) A-IV; B-I; C-II; D-III
(4) A-IV; B-II; C-I; D-III

71. The sum of total number of protons and neutrons in the isotopes of hydrogen is:
(1) 9 (2) 3
(3) 8 (4) 6

72. How many among the following are temperature independent terms?
Density, Molality, Molarity, Mole fraction, % by (mass/volume), % by (mass/mass)
(1) 3 (2) 4
(3) 2 (4) 5

73. The wavelength of an electron having kinetic energy 9.1×10^{-25} J will be:
(Given: $m_e = 9.1 \times 10^{-31}$ kg and $h = 6.62 \times 10^{-34}$ Js)
(1) 0.84×10^{-5} m
(2) 0.51×10^{-6} m
(3) 0.71×10^{-7} m
(4) 0.26×10^{-8} m

74. Ebullioscopic constant is the elevation in boiling point produced by:
(1) 1% (mass/mass) solution
(2) 1% (mass/volume) solution
(3) 1 molar solution
(4) 1 molal solution

75. **Incorrect** statement among the following is/are:
(A) The energies of the orbitals in the same subshell decrease with increase in the atomic number (Z_{eff}).
(B) The ground state electronic configuration of the atom of an element always corresponds to the state of lowest total electronic energy.
(C) All four quantum numbers (n, l, m_l and m_s) can be obtain by solving Schrodinger wave equation.
(D) The probability of finding an electron at a point within an atom is proportional the square of the orbital wave function.
(1) A and C only
(2) B and D only
(3) A and B only
(4) C only

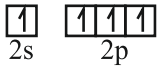
76. When a non-volatile solute is added to the solvent, the vapour pressure of the solvent decreases by 10 mm of Hg. The mole fraction of the solute in the solution is 0.2. What will be the mole fraction of the solvent if decrease in vapour pressure is 5 mm of Hg?
(1) 0.85 (2) 0.80
(3) 0.95 (4) 0.90

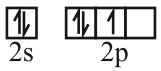
77. If an electron in H-atom is at a distance of 2.116 Å from the nucleus then its angular momentum is:
(1) $\frac{h}{2\pi}$
(2) $\frac{h}{\pi}$
(3) $\frac{3h}{2\pi}$
(4) $\frac{2h}{\pi}$

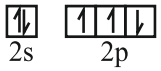
78. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Aquatic species are more comfortable in cold water rather than in warm water.
Reason R: On increasing temperature solubility of oxygen in water decreases.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

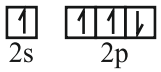
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79. The orbital diagram in which both the Aufbau principle and Hund's rule are violated, is:

(1) 

(2) 

(3) 

(4) 

80. The percentage by mass of chloroform in a sample of drinking water containing 10 ppm (by mass) of chloroform is:

(1) 1×10^{-3} %

(2) 1×10^{-6} %

(3) 1×10^{-4} %

(4) 1×10^{-5} %

81. With increase of positive charge of the nucleus and principal quantum number, the kinetic energy of electron in one electron species:

(1) increases and decreases respectively

(2) decreases and increases respectively

(3) increases

(4) decreases

82. The mass percentage of the resulting solution obtained by mixing 100 g of 50% solution and 200 g of 60% solution by mass is:

(1) 52.43%

(2) 56.66%

(3) 64.21%

(4) 48.22%

83. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:

Assertion A: There is no rigid size boundary surface diagram, which bounds a region in which the probability of finding the electron is 100%.

Reason R: The probability density $|\psi|^2$ has always some value, howsoever small it may be, at infinite distance from the nucleus.

In the light of above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

84. **Incorrect** statement among the following is:

(1) Naphthalene dissolve readily in benzene.

(2) Ethylene glycol is used as antifreeze compound in car engines.

(3) Osmosis is used in desalination of sea water.

(4) van't Hoff factor is the ratio of normal molar mass to experimentally determined molar mass.

85. The number of radial nodes and angular nodes in 4s orbital respectively are:

(1) 4 and 0 (2) 4 and 3

(3) 3 and 0 (4) 3 and 2

86. 0.1 molal aqueous solution of the complex $CrCl_3 \cdot xNH_3$ shows a depression in freezing point of 0.372°C. Assuming 100% ionisation of this complex, the complex may be:

(Given: K_f of H_2O = 1.86 K kg mol^{-1})

(1) $[Cr(NH_3)_6]Cl_3$

(2) $[Cr(NH_3)_5Cl]Cl_2$

(3) $[Cr(NH_3)_4Cl_2]Cl$

(4) $[Cr(NH_3)_3Cl_3]$

87. Which among the following orbitals does **not** have electron density along the axis?

(1) d_{z^2} (2) $d_{x^2-y^2}$

(3) d_{xy} (4) p_y

88. A 1% (w/v) solution of Urea (Molar mass = 60 g mol^{-1}) is isotonic with 2% (w/v) solution of an unknown solute. The molar mass of unknown solute (in g mol^{-1}) is:

(1) 120 (2) 180

(3) 90 (4) 240

89. A microscope using suitable photons is employed to locate an electron in an atom within a distance of 1 Å. The uncertainty involved in the measurement of the velocity is nearly:

(1) 5.8×10^5 ms^{-1}

(2) 3.2×10^6 ms^{-1}

(3) 2.1×10^4 ms^{-1}

(4) 7.6×10^7 ms^{-1}

90. Which of the following binary mixture forms minimum boiling azeotropes?

(1) Ethanol + Water

(2) Aniline + Phenol

(3) Benzene + Toluene

(4) Chloroform + Acetone

[Contd...

91. Select the correct statements regarding the phase of the cell cycle characterised by duplication of genetic material in eukaryotic cells.

A. The cells in this phase are metabolically inactive but are preparing to divide.
B. The amount of DNA in the cell doubles.
C. In plant cells, centriole also duplicates in the nucleus.
D. After this phase, the cell will have same number of chromosomes as present in its previous phase.
E. This phase is not a part of meiosis.

In the light of above statements choose the **correct** answer from the options given below.

(1) A, B and C are incorrect
(2) B and D are correct
(3) A, C and E are correct
(4) A, B and E are incorrect

92. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

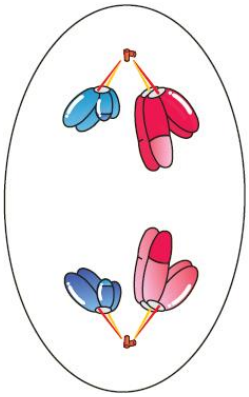
Assertion (A): Cytoplasmic increase in a cell occurs intermittently during cell division.

Reason (R): The events of cell growth and its division are under genetic control.

In the light of the above statements, choose the *most appropriate* from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

93. Go through the stage of cell cycle shown in the following diagram.



Select the **correct** option regarding it.

(1) The chromosomes are getting ready for the exchange of genetic material.
(2) After this stage, the bivalents of homologous chromosomes will be clearly visible.
(3) The separated homologous chromosomes are moving towards the opposite poles.
(4) This phase is the third stage of M phase in equational division.

94. Arrange the following details/events of Prophase I in the correct sequence of their occurrence.

A. The homologous chromosomes remain attached only at the sites of crossing over.
B. The enzyme recombinase is actively catalysing the process of crossing over.
C. The homologous chromosomes come together and form pairs.
D. Chromatin is completely condensed into chromosomes.
E. Compaction of chromosomes starts so that they can be gradually visualised under the light microscope.

Choose the **correct** sequence from the options below.

(1) E → C → B → A → D
(2) C → E → B → A → D
(3) E → B → C → A → D
(4) E → C → A → B → D

95. How many of the following statements about the G_0 phase are **correct**?

A. All human cells are found in this phase of cell cycle.
B. These cells can re-enter the cell cycle via the G_1 phase.
C. Usually, these cells are metabolically inactive that leads to their death.
D. The cells exit the cell cycle after meiosis I to enter this stage.
E. This phase is not seen in diploid cells.

Choose the **correct** option.

(1) One only
(2) Two only
(3) Three only
(4) Four only

96. Given below are two statements:
Statement I: Meiosis ensures the production of a haploid phase in the life cycle of sexually reproducing organisms.
Statement II: Mitosis restores the diploid phase in the life cycle of sexually reproducing organisms.
In the light of the above statements, choose the **correct** from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

97. Suppose the gamete of a male honey bee has 'N' number of chromosomes. How many bivalents will be formed in it?
(1) N
(2) Zero
(3) 2N
(4) N/2

98. A cell enters the G_1 phase of cell cycle with 32 chromosomes. After S phase the number of pairs of sister chromatids in it will be;
(1) 16
(2) 32
(3) 64
(4) 8

99. Which of the following pairs is **correctly** matched?
(1) Meiotic prophase I - Longer than meiotic prophase II
(2) Meiotic telophase I - Shows disintegration of nucleolus unlike mitotic telophase
(3) Meiotic anaphase I - Similar to mitotic anaphase
(4) Meiotic metaphase I - Same number of chromosomes as in metaphase II

100. Arrange the following in descending order of their respective durations.
A. Cell cycle in yeast.
B. Interphase in a human cell in culture.
C. M phase in yeast.
D. Cell cycle in a human cell in culture.
E. Interphase in yeast.
F. M phase in a human cell in culture.
Choose the *most appropriate* option from below.
(1) D > B > F > A > E > C
(2) B > D > A > E > F > C
(3) D > B > A > E > F > C
(4) D > B > A > F > E > C

101. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): By undergoing mitosis, a cell having 16 chromosomes will produce two daughter cells having 8 chromosomes each.
Reason (R): Mitosis is said to be equational division i.e. the chromosomes of parent cell are equally distributed among the daughter cells.
In the light of the above statements, choose the *most appropriate* from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

102. Match **List-I** with **List-II**.

List-I (Description)		**List-II (Number of stages divided into)**	
(A)	Resting phase of cell cycle	(I)	Three
(B)	The most dramatic phase of cell cycle	(II)	Five
(C)	First stage of meiosis I	(III)	Two
(D)	Overall cell cycle	(IV)	Four

Choose the *most appropriate* from the options given below.
(1) A-IV, B-I, C-II, D-III
(2) A-IV, B-II, C-I, D-III
(3) A-III, B-I, C-II, D-IV
(4) A-I, B-IV, C-II, D-III

103. Choose the **correct** statement about liquid endosperm of a coconut.
(1) Each nucleus is present in distinct cellular entity surrounded by cell wall.
(2) Produced when nuclear division is not followed by cytoplasmic division.
(3) Its condition is opposite to the cellular endosperm in the kernel in which syncytium is seen.
(4) It is usually enucleated and hence short-lived.

104. A scientist viewing a human cell under microscope to study the events of cell cycle made following observations over a period of time:

A. He was able to study the size and shape of the chromosomes arranged on a single equatorial plate.
B. Out of four chromatids, two sets of two chromatids each were formed.
C. A cluster of four cells was formed.
D. The nuclear membrane was intact and a non-membranous cytoplasmic organelle was duplicating.
E. Two cells were seen without any major event of karyokinesis or cytokinesis.

Choose the **correct** order of sequence of these events.

(1) D → A → B → E → C
(2) A → D → B → E → C
(3) D → B → A → E → C
(4) D → B → E → A → C

105. Given below are two statements:

Statement I: The reduction division involves two sequential cycles of cytoplasmic division, and one cycle each of nuclear division and DNA replication.

Statement II: Identical sister chromatid for each chromosome is synthesised at the S phase of cell cycle.

In the light of the above statements, choose the **correct** from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

106. Microscopic examination of a plant tissue treated with an experimental drug reveals several large cells, each possessing two genetically identical, diploid nuclei. This drug most likely has a mechanism that inhibits the:

(1) polymerization of tubulin proteins to form spindle fibers.
(2) activity of the enzyme responsible for crossing over.
(3) formation of the precursor of future cell wall of daughter cell.
(4) activity of DNA synthesising enzyme before the start of M-phase.

107. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Interphase is called the resting phase of cell cycle.

Reason (R): In interphase, the cell is virtually in a state of relaxation wherein it is metabolically not much active and has no role in the division of the cell.

In the light of the above statements, choose the *most appropriate* from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

108. Identify the phases of cell cycle based on the given description.

A. Stage that occurs only in meiosis and is short lived.
B. The interval between mitosis and initiation of DNA replication.
C. The stage which may last for months or years in oocytes of some vertebrates
D. Stage of meiosis I when spindle apparatus assembles.

Choose the **correct** option.

(1) A – Diplotene; B – Interkinesis; C – G_1 phase; D – Diakinesis
(2) A – Diakinesis; B – Diplotene; C – Interkinesis; D – G_1 phase
(3) A – G_1 phase; B – Diakinesis; C – Interkinesis; D – Diplotene
(4) A – Interkinesis; B – G_1 phase; C – Diplotene; D – Diakinesis

109. A hypothetical tetraploid (4n) plant cell in the G_1 phase has a DNA content of 4C. Match the description/stage in **List-I** with the correct number of chromosomes and amount of DNA per cell in **List-II**.

List-I		**List-II**	
(A)	First stage of meiosis I	(I)	4n chromosomes and 4C DNA
(B)	Stage between meiosis I and meiosis II	(II)	2n chromosomes and 4C DNA
(C)	After cytokinesis of meiosis II	(III)	4n chromosomes and 8C DNA
(D)	Start of S-phase	(IV)	2n chromosomes and 2C DNA

Choose the *most appropriate* from the options given below.

(1) A-IV, B-I, C-II, D-III
(2) A-III, B-II, C-IV, D-I
(3) A-III, B-I, C-II, D-IV
(4) A-I, B-IV, C-II, D-III

110. Examine the following statements related to the cell cycle and identify the **correct** ones.

A. The sequence of events by which a cell duplicates its genome and eventually divides into two is called the cell cycle.
B. Sexually reproducing small organisms start life from a single-celled zygote, while the larger organisms start it from a multi-celled zygote.
C. According to the cell theory, cells arise from pre-existing cells.
D. Cell division, DNA replication, and cell growth must occur in a coordinated way.
E. Cell cycle starts with division of cytoplasm.

Choose the *most appropriate* option from below.

(1) A, B and D
(2) A, C and D
(3) B, D and E
(4) A, C and E

111. Identify the **incorrect** statement for the phase of cell cycle in which virtually all the components of the cell are reorganised.

(1) It is called the equational division.
(2) It starts with the decondensation of chromatin and ends with its condensation.
(3) It is the most dramatic period of cell cycle.
(4) For convenience it has been divided into four stages of karyokinesis.

112. Meiosis II is often compared to mitosis. Which of the following events is a key feature that makes Meiosis II similar to mitosis?

(1) Pairing of homologous chromosomes
(2) Separation of homologous chromosomes
(3) Separation of sister chromatids
(4) Reduction of chromosome number to half

113. Identify the **correct** statements regarding the significance of mitosis.

A. It leads to the growth of multicellular organisms.
B. It disturbs the nucleo-cytoplasmic ratio.
C. It acts significantly in replacing cells of the gut lining.
D. In some lower plants, it is the process adopted for gamete formation.
E. In plants, it contributes to continuous growth throughout life by division in permanent tissues.

Choose the *most appropriate* option.

(1) A, C, D
(2) B, C, D
(3) B, C, E
(4) A, D, E

114. A cell exited the cell cycle and entered the G_0 phase. Identify the **correct** statement for the stage that it has exit?

(1) It is also called the quiescent stage.
(2) The cell is metabolically inactive.
(3) The cell is actively synthesising proteins required for DNA replication.
(4) The cell is about to undergo cytoplasmic division.

115. Match the event in **List-I** with the corresponding phase of meiosis in **List-II**.

List-I		List-II	
(A)	Formation of bivalents	(I)	Anaphase II
(B)	Terminalisation of chiasmata	(II)	Zygotene
(C)	Splitting of centromeres	(III)	Pachytene
(D)	Appearance of recombination nodules	(IV)	Diakinesis

Choose the *most appropriate* from the options given below.

(1) A-IV, B-II, C-I, D-III
(2) A-II, B-IV, C-I, D-III
(3) A-II, B-III, C-I, D-IV
(4) A-III, B-IV, C-I, D-II

116. Choose the **incorrectly** matched pair.

(1) Mitotic apparatus: Forms by association of asters and spindle fibres
(2) Spindle fibres : A type of microtubules
(3) Kinetochores : Bind to the centrioles from one pole
(4) Asters : Radiated out from centrosomes in animal cells

117. Which of the following statements about cytokinesis are **correct**?

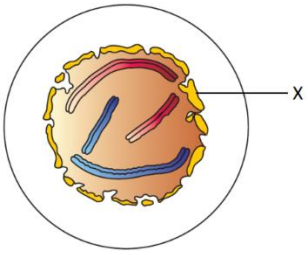
A. In animal cells, it is achieved by the appearance of a furrow in the wall.
B. In plant cells, wall formation starts in the periphery and grows inward.
C. The cell plate represents the middle lamella between the walls of adjacent cells.
D. Mitochondria and plastids get distributed between the two daughter cells during this process.
E. It is followed by division of the nucleus.

Choose the most appropriate option.

(1) C and D only
(2) B and E only
(3) A, B and C only
(4) A and D only

[Contd...

118. The following diagram represents a stage of cell cycle. Choose the correct statement about it.



(1) This is the first stage of karyokinesis in a plant cell.
(2) The chromatin is compacting but is yet to be fully condensed.
(3) The complete integration of structure X marks the start of second phase of M phase.
(4) The chromosomes soon reach an extremely extended state.

119. Given below are two statements:
Statement I: The gametes of sexual reproduction can be diploid only.
Statement II: The zygote formed during sexual reproduction can be haploid or diploid.
In the light of the above statements, choose the **correct** from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

120. A student is observing a human cell undergoing cell division wherein unequal sized cells are produced after its completion. What is most likely to be the correct description regarding this?
(1) He is observing the male germ cells.
(2) He is observing the female germ cells.
(3) The cells are undergoing meiotic division that has not completed in the correct manner.
(4) These all cells will be genetically identical.

121. If a cell has 6n chromosomes, then the number of chromosomes in its gamete and zygote, respectively, will be;
(1) 3n and 12n (2) 6n and 6n
(3) 6n and 12n (4) 3n and 6n

122. A pair of chromatids attached to common centromere appearing to move towards opposite poles will;
(1) have the centromere directed towards the equatorial plate.
(2) be pulled due to shortening of microtubules.
(3) be present in the second nuclear division of reduction division.
(4) separate in the very next stage.

123. Identify the correct sequence of events that occurs during mitotic cell cycle.
A. Proteins are synthesised in preparation for mitosis.
B. Centrosomes move towards opposite poles.
C. Nucleolus, golgi complexes and ER reappear.
D. Chromosomes align on the equatorial plate.
E. Daughter chromosomes are formed.
The **correct** order of events is:
(1) A → B → D → E → C
(2) A → E → B → D → C
(3) E → A → B → C → D
(4) B → A → E → D → C

124. Given below are two statements:
Statement I: The complex formed by a pair of synapsed homologous chromosomes is called a bivalent, which is more clearly visible in the pachytene stage than the previous stage.
Statement II: The synapsed chromosomes separate after crossing over except at the sites of crossing over, in diplotene.
In the light of the above statements, choose the **correct** from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

125. After S phase, the number of which of the following is doubled?
(1) Cell
(2) Chromosomes
(3) Chromatids
(4) Nucleus

126. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): At the end of Anaphase I of meiosis, each pole receives double the chromosome number of the parent cell.
Reason (R): The DNA amount of cell is double than that of parent cell in this stage.
In the light of the above statements, choose the *most appropriate* from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

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127. Which of the following statements about Meiosis is/are incorrect?
A. Also called reduction division.
B. Mainly required for repair or replacement of injured/dead cells.
C. DNA replication occurs during interphase.
D. Four haploid cells are formed at the end of Meiosis II.
E. Always leads to variations in progeny.
Choose the *most appropriate* option.
(1) B only (2) B and C only
(3) D and E only (4) A and B only

128. In the anaphase I of a cell, 12 chromosomes were moving towards one pole and 12 chromosomes towards the other pole. The correct number of chromosomes in its gamete will be;
(1) 6 (2) 12
(3) 18 (4) 24

129. Match **List-I** with **List-II**.

List-I (Event of cell cycle)		**List-II (Simultaneous event)**	
(A)	DNA synthesis	(I)	Centriole movement to opposite poles
(B)	Chromosomal condensation	(II)	Centriole duplication
(C)	Nuclear membrane reformation	(III)	Distribution of mitochondria in daughter cells
(D)	Cell furrow formation	(IV)	Golgi and ER reappear

Choose the *most appropriate* from the options given below.
(1) A-I, B-III, C-IV, D-II
(2) A-I, B-IV, C-III, D-II
(3) A-II, B-I, C-IV, D-III
(4) A-IV, B-I, C-II, D-III

130. Consider the following statements about the mitotic stage that is followed by cytokinesis.
A. Chromosomes lose their individuality.
B. Cell becomes metabolically inactive.
C. A nuclear envelope assembles around the chromosome clusters.
D. Cell exits the cell cycle.
E. Chromatin material collects at the centre of the cell.
Identify the option with **incorrect** statements.
(1) A, C and D (2) B, D and E
(3) A, B and C (4) C, D and E

131. The duration of the cell cycle:
(1) is the same for all prokaryotic and eukaryotic cells.
(2) is approximately 24 minutes for a typical human cell.
(3) can vary from organism to organism and also from cell type to cell type.
(4) is a static parameter for all living species.

132. If one kinetochore of a chromosome during metaphase of mitosis fails to attach to spindle fibres, what would be the most likely immediate consequence in the subsequent stage?
(1) Both sister chromatids of that chromosome would move to the same pole.
(2) The centromere of the chromosome would fail to divide, and the entire chromosome would lag at the equator.
(3) One of the daughter chromatids would fail to migrate to a pole, while its sister moves to the opposite pole.
(4) The cell would immediately exit the cell cycle and enter the quiescent stage.

133. The fundamental difference that ensures a diploid parent cell produces diploid daughter cells in mitosis, but a diploid parent cell produces haploid cells after meiosis I, is the:
(1) Duplication of centrioles in the cytoplasm during the S phase.
(2) Condensation of chromosomal material during prophase.
(3) Alignment and subsequent separation of chromosomes during metaphase and anaphase of mitosis and that of meiosis I.
(4) Reformation of the nuclear envelope during telophase.

134. Match **List-I** with **List-II**.

List-I		**List-II**	
(A)	Bivalent	(I)	Multinucleate condition
(B)	Syncytium	(II)	Hold the sister chromatids together
(C)	Chiasmata	(III)	Also called tetrad
(D)	Centromere	(IV)	X-shaped structures

Choose the *most appropriate* from the options given below.
(1) A-III, B-I, C-IV, D-II
(2) A-III, B-II, C-IV, D-I
(3) A-II, B-I, C-III, D-IV
(4) A-IV, B-I, C-II, D-III

135. Paired homologous chromosomes which are yet to exchange genetic material are said to be present in between;
(1) meiosis I and meiosis II.
(2) telophase I and prophase II.
(3) pachytene and diakinesis.
(4) zygotene and pachytene.

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136. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: An individual with blood group O can donate blood to individuals with blood groups A, B, and AB.
Reason R: The plasma of an individual with blood group O contains both anti-A and anti-B antibodies.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

137. What is the primary function of the thin layer of pleural fluid found between the visceral and parietal pleurae?
(1) To maintain a positive pressure that forces the lungs to expand.
(2) To act as a lubricant, allowing the lungs to glide smoothly against the thoracic wall during breathing.
(3) To facilitate the diffusion of gases between the lungs and the chest wall.
(4) To provide a medium for the transport of nutrients to the lung tissue.

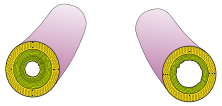
138. How many of the following statements are **correct**?
A. The SA node generates the maximum number of action potentials per minute, so it sets the pace of the activities of the heart.
B. The stroke volume is approximately 70 ml i.e., blood pumped out by each ventricle during a cardiac cycle.
C. The heart is auto excitable, and its functions can be moderated by neural and hormonal mechanisms.
D. The systemic heart includes the right ventricle and left atrium.
(1) One (2) Two
(3) Three (4) Four

139. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The pO_2 of alveolar air is always lower and the pCO_2 is always higher than that of inspired atmospheric air.
Reason R: This is because the process of gas exchange is continuous; oxygen is constantly diffusing out of the alveoli into the blood, while carbon dioxide is constantly diffusing from the blood into the alveoli.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

140. Given below are two statements.
Statement I: QRS complex of an ECG helps in determining heart rate.
Statement II: Mitral valve is present between left ventricle and left atrium.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

141. The rate of diffusion of gases across the alveolar membrane is **not** dependent on which of the following factors?
(1) The thickness of the diffusion membrane.
(2) The partial pressure gradient of the gas.
(3) The concentration of carbonic anhydrase in RBCs.
(4) The solubility of the gas in the fluid medium.

142. Based on the provided diagram of blood vessels, among the following statement, choose the statement that is **correct**.



A **B**

(1) The pressure of blood within the **A** connected to the right ventricle is higher than the pressure in the **A** connected to the left ventricle.
(2) The **B** that enters the left atrium carries blood with a lower concentration of oxygen than the **B** that enters the right atrium.
(3) The wall of the **A** leaving the heart for the systemic circuit is thicker and more elastic than the wall of the **B** returning to the heart from the systemic circuit.
(4) All vessels labelled **A** carry oxygenated blood, and all vessels labelled **B** carry deoxygenated blood.

143. Given below are two statements.

Statement I: The right lung is divided into two lobes by a single oblique fissure, while the left lung is divided into three lobes by horizontal and oblique fissures.

Statement II: The blood-air barrier, which facilitates rapid gaseous exchange in the lungs, is a composite structure formed by the simple squamous epithelium of the alveolus, the endothelium of the pulmonary capillary, and their fused basement membranes.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

144. Match **List-I** with **List-II**.

List-I		**List-II**	
(A)	Blood	(I)	55% of blood
(B)	Lymph	(II)	Most commonly used body fluid for the transport of substances
(C)	Coelenterates	(III)	Helps in absorption of fats
(D)	Plasma	(IV)	Lacks a circulatory system

Choose the **correct** answer from the options given below:

(1) A-II, B-I, C-III, D-IV
(2) A-II, B-III, C-IV, D-I
(3) A-I, B-II, C-III, D-IV
(4) A-II, B-IV, C-I, D-III

145. What is the primary effect of a strong signal from the pneumotaxic centre on regulation of respiration?

(1) It increases the duration of inspiration, leading to a decrease in respiratory rate.
(2) It has no effect on inspiration but increases the force of expiration.
(3) It reduces the duration of inspiration, leading to an increase in respiratory rate.
(4) It completely inhibits the respiratory rhythm centre.

146. A woman with B-negative blood type is pregnant for the second time. Her first child was born healthy with an O-positive blood type. The father of both children has AB-positive blood. Which of the following statements most **accurately** describes the potential risk of *erythroblastosis fetalis* in her second pregnancy?

(1) There is no risk, because the mother's first child had a different ABO blood group, which prevents Rh sensitization.
(2) There is a significant risk of a severe reaction if the second fetus is Rh-positive, as the mother was likely sensitized to the Rh antigen during her first delivery.
(3) The greatest risk is due to ABO incompatibility, where the mother's anti-A antibodies will attack the red blood cells of an A-positive or AB-positive fetus.
(4) There is no risk, because the condition only occurs when the mother is Rh-positive and the fetus is Rh-negative.

147. Match **List-I** with **List-II**.

List-I (Animal)		**List-II (Respiratory organs)**	
(A)	Earthworm	(I)	Lungs
(B)	Aquatic Arthropod	(II)	Network of tracheal tubes
(C)	Terrestrial Reptile	(III)	Moist Cuticle
(D)	Cockroach	(IV)	Gills

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-I, D-II
(2) A-III, B-I, C-IV, D-II
(3) A-II, B-IV, C-I, D-III
(4) A-IV, B-III, C-II, D-I

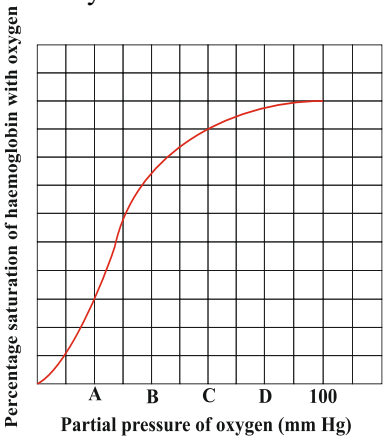
148. Consider a population with B-positive blood. How many of the following statements regarding their blood are **true**?

A. The surface of their erythrocytes possesses both B and Rh antigens.
B. Their blood plasma lacks anti-A antibody.
C. They are considered a universal recipient for plasma transfusions.
D. A transfusion of their blood into an A-positive individual would likely cause a haemolytic reaction.

(1) Only one statement is true.
(2) Only two statements are true.
(3) Only three statements are true.
(4) All four statements are true.

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149. In the given diagram of the oxygen dissociation curve, under normal physiological conditions, the partial pressure of oxygen at the tissue level is represented by:



(1) A (2) B
(3) C (4) D

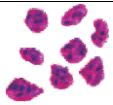
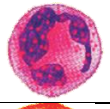
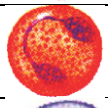
150. A medical student is comparing samples of blood and lymph taken from a healthy individual. Which of the following observations would be the most **accurate** and fundamental difference between the two fluids?

(1) The lymph sample contains a higher concentration of plasma proteins, such as albumin and fibrinogen, than the blood sample.
(2) The blood sample is red due to the presence of erythrocytes for oxygen transport, while the lymph sample is a clear, colourless fluid lacking erythrocytes.
(3) Both fluids contain a similar number of leukocytes, but the lymph sample is unique in that it also contains a significant number of platelets for clotting.
(4) The primary function of blood is immunity, whereas the primary function of lymph is to transport nutrients and hormones throughout the body.

151. What is the primary cause of emphysema?

(1) Cigarette smoking
(2) Bacterial infection of the lungs
(3) Allergic reaction to dust
(4) Inhaling industrial pollutants

152. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Eosinophil	(I)	
(B)	Thrombocytes	(II)	
(C)	T lymphocyte	(III)	
(D)	Neutrophil	(IV)	

Choose the **correct** answer from the options given below:

(1) A-IV, B-I, C-III, D-II
(2) A-III, B-II, C-I, D-IV
(3) A-III, B-I, C-IV, D-II
(4) A-IV, B-III, C-I, D-II

153. How many of the following events occur during a forceful expiration?

A. Contraction of the diaphragm.
B. Contraction of external intercostal muscles.
C. Contraction of abdominal muscles.
D. An increase in intra-abdominal pressure.
E. Contraction of internal intercostal muscles.

(1) Two
(2) Three
(3) Four
(4) One

154. Given below are two statements.

Statement I: A key evolutionary advancement of the four-chambered heart is its ability to maintain two completely separate circulatory pathways, ensuring that systemic tissues receive fully oxygenated blood.

Statement II: Compared to open circulatory systems, closed systems provide the advantage of rapid and targeted blood delivery, allowing for more efficient response to the varying oxygen requirements of different tissues.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

155. Identify the **correct** sequence of structures in the human respiratory tract, starting from the outside.

(1) Pharynx → Nasal Chamber → Larynx → Trachea → Bronchi → Bronchioles → Alveoli
(2) Nasal Chamber → Pharynx → Larynx → Trachea → Bronchi → Bronchioles → Alveoli
(3) Nasal Chamber → Larynx → Pharynx → Trachea → Bronchioles → Bronchi → Alveoli
(4) Pharynx → Larynx → Trachea → Nasal Chamber → Bronchi → Bronchioles → Alveoli

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156. How many of the following statements that compare and contrast erythrocytes (RBCs) and leucocytes (WBCs) are **correct**?

A. Erythrocytes are the most numerous cells in the blood, while leucocytes are relatively lesser in number.
B. A key structural difference is that mature RBCs in most mammals lack a nucleus, whereas WBCs are nucleated.
C. The presence of haemoglobin makes RBCs red and functional in gas transport, while the absence of this pigment makes WBCs colourless.
D. Both cell types are considered short-lived, with RBCs being destroyed in the spleen after an average of 120 days.

(1) One
(2) Two
(3) Three
(4) Four

157. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): While swallowing food can't enter the larynx.
Reason (R): During swallowing the opening of larynx, glottis is covered by epiglottis.
In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is false and R is true.
(2) Both A and R are true and R is a correct explanation of A.
(3) Both A and R are true but R is not a correct explanation of A.
(4) A is true and R is false.

158. Which of the following **correctly** defines the peaks of a standard ECG.

	P-wave	QRS complex	T-wave
(1)	Depolarisation of the atria	Repolarisation of ventricles	Depolarisation of ventricles
(2)	Depolarisation of atria	Depolarisation of the ventricles	Repolarisation of ventricles
(3)	Repolarisation of the ventricles	Repolarisation of atria	Depolarisation of the ventricles
(4)	Depolarisation of the ventricle	Depolarisation of the atria	Repolarisation of the atria

159. Which of the following pairs representing a respiratory volume/capacity and its standard value is **incorrectly** matched?

A. Inspiratory Capacity (IC) : 3500 mL
B. Functional Residual Capacity (FRC) : 2300 mL
C. Vital Capacity (VC) : 5800 mL
D. Total Lung Capacity (TLC) : 5800 mL

(1) Only C (2) Only D
(3) Both A and C (4) Both C and D

160. Choose the option that **correctly** identifies the pair(s) as correct or incorrect.

(A)	Hypertension	(I)	Transient high blood pressure
(B)	Hepatic portal system	(II)	Hepatic portal vein carries blood from liver to the intestine
(C)	Coronary circulation	(III)	Provides blood circulation to cardiac musculature
(D)	Pericardium	(IV)	Muscular protective covering of heart

(1) Only pair B–II is incorrect
(2) Only pair C–III is incorrect
(3) Only pair A–I, B-II and D–IV are incorrect
(4) All pairs are correct

161. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Spirometer	(I)	Cannot measure residual volume
(B)	Inspiration	(II)	Diaphragm flattens
(C)	Pharynx	(III)	Have cartilaginous rings
(D)	Secondary bronchioles	(IV)	Common route for air and food

Choose the **correct** answer from the options given below:

(1) A-I, B-III, C-II, D-IV
(2) A-IV, B-I, C-III, D-II
(3) A-I, B-II, C-III, D-IV
(4) A-I, B-II, C-IV, D-III

162. What is the most critical purpose of the brief delay at the AV node after it receives an impulse from the SA node?

(1) To allow the atria and ventricles to contract simultaneously.
(2) To allow the ventricles sufficient time to fill with blood.
(3) To increase the overall heart rate set by the SA node.
(4) To amplify the electrical signal before it reaches the ventricles.

163. Which of the following breathing patterns is the most efficient for maximizing gas exchange in the alveoli?

(1) Increasing the respiratory rate while decreasing the tidal volume.
(2) Decreasing the respiratory rate while increasing the tidal volume.
(3) Increasing both the respiratory rate and the tidal volume to their maximums.
(4) Maintaining a constant pulmonary ventilation rate, regardless of the breathing pattern.

164. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The complexity and design of circulatory systems, from simple diffusion to complex multi-chambered hearts, vary greatly across the animal kingdom.
Reason R: To maintain homeostasis and ensure survival, only few living cell within an organism must be supplied with necessary resources and have its metabolic byproducts cleared away.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

165. During a period of strenuous physical activity, which of the physiological responses listed below are expected to occur?
A. An increase in the partial pressure of O_2 in the tissues.
B. A decrease in the pH of the blood due to lactic acid and CO_2 accumulation.
C. Vasoconstriction of blood vessels supplying skeletal muscles to conserve energy.
D. An increase in the rate and depth of breathing.
E. A rightward shift in the oxygen-haemoglobin dissociation curve.
(1) B, D and E (2) A, B and D
(3) A, C and E (4) C, D and E

166. During an acute "fight-or-flight" response, both direct sympathetic nerve stimulation and the release of adrenaline from the adrenal medulla act on the heart. What is the resulting effect on cardiac function?
(1) The two systems have an antagonistic effect; the direct neural stimulation increases heart rate, while adrenaline primarily decreases it, leading to an unstable rhythm.
(2) A powerful, synergistic effect occurs, causing a rapid and sustained increase in both the heart rate and the force of contraction, leading to a maximal increase in cardiac output.
(3) Only the direct sympathetic nerve stimulation has an immediate effect on heart rate; the hormonal effect of adrenaline is too slow and primarily serves to constrict blood vessels in the periphery.
(4) The heart rate increases significantly, but the force of ventricular contraction decreases to prevent over-exertion of the cardiac muscle, resulting in a minimal change to overall blood flow.

167. Which of the following is the most potent and primary chemical stimulus for regulating the rate and depth of breathing in a healthy human at rest?
(1) The partial pressure of oxygen (pO_2) in arterial blood.
(2) The partial pressure of carbon dioxide (pCO_2) in arterial blood.
(3) The concentration of bicarbonate ions (HCO_3^-) in the blood.
(4) The partial pressure of nitrogen (pN_2) in alveolar air.

168. Compared to a healthy but sedentary individual, a highly trained marathon runner at rest would exhibit which of the following characteristics in their cardiac cycle?
(1) A significantly higher resting cardiac output, achieved by a very rapid heart rate.
(2) A lower resting heart rate and a significantly increased stroke volume
(3) A shorter overall cardiac cycle duration, with a reduced time for ventricular filling to make the heart more efficient.
(4) An identical stroke volume but a much faster ventricular systole, allowing more cycles to be completed per minute.

169. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Normal breathing rate	(I)	Sound box
(B)	Epiglottis	(II)	Thin elastic cartilaginous flap
(C)	100 ml oxygenated blood	(III)	12-16 times/minute.
(D)	Larynx	(IV)	Can deliver around 5 ml of O_2 to the tissues under normal physiological conditions.

Choose the **correct** answer from the options given below:
(1) A-IV, B-III, C-II, D-I
(2) A-III, B-II, C-IV, D-I
(3) A-III, B-IV, C-I, D-II
(4) A-III, B-IV, C-II, D-I

170. Select the option that **correctly** arranges the following events in the sequential order they occur during blood coagulation.

A. Formation of an enzyme complex called thrombokinase.
B. Conversion of inactive fibrinogen into a network of fibrin threads.
C. Activation of the clotting mechanism by factors released from platelets or injured tissue.
D. Formation of the enzyme thrombin from its inactive precursor in the plasma.
E. Trapping of dead and damaged formed elements to form a coagulum.

(1) C → A → D → B → E
(2) C → D → A → B → E
(3) A → C → D → E → B
(4) C → A → B → D → E

171. Which of the following pairs of locations in the human body is characterized by a partial pressure of oxygen (pO_2) of approximately 95 mmHg and a partial pressure of carbon dioxide (pCO_2) of approximately 40 mmHg?

(1) Air in the alveoli and blood in the aorta
(2) Systemic tissues and blood in the pulmonary artery
(3) Blood in the pulmonary vein and blood in the systemic arteries
(4) Expired air and blood in the systemic veins

172. Blood is classified as a connective tissue because it is composed of which two fundamental components?

(1) A cellular component (leukocytes) and a solid extracellular matrix.
(2) A fluid matrix (serum) and dissolved protein fibers only.
(3) Cellular components (formed elements) suspended within a fluid extracellular matrix (plasma).
(4) A cellular matrix made of plasma and suspended red blood cells.

173. Which of the following statements regarding the transport of carbon dioxide are **true**?

A. The majority (about 70%) of CO_2 is transported in a dissolved state in the plasma.
B. The enzyme carbonic anhydrase is present in high concentrations in RBCs and in minute quantities in plasma.
C. The formation of carbamino-haemoglobin is primarily influenced by the partial pressure of O_2.
D. Every 100 ml of deoxygenated blood delivers approximately 4 ml of CO_2 to the alveoli.

(1) A and C (2) B and D
(3) A and D (4) B and C

174. Which of the following statements provides the most **accurate** functional description of the structures mentioned?

(1) Chordae tendineae works to actively pull the atrioventricular (AV) valves open, allowing blood to enter the ventricles.
(2) Valves between right atrium and right ventricle has three muscular flaps similar to the three cusps present in the valve between left atrium and left ventricle.
(3) During ventricular contraction, the papillary muscles contract, tensing the chordae tendineae to prevent the cusps of the AV valves from being forced back into the atria.
(4) The primary role of both the valves within the heart and the valves within the veins is to ensure that oxygenated and deoxygenated blood do not mix.

175. Given below are two statements.

Statement I: Lub and Dub are the first and second sounds of heart produced during consecutive cardiac cycles.

Statement II: The apex of heart is the pointed end of the heart directed downwards and to the left.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

176. Which of the following statements are **correct**?

A. The condition known as atherosclerosis is characterized by the narrowing of arteries due to deposits of materials like calcium and cholesterol.
B. Angina pectoris is another name for a heart attack, where the heart muscle is suddenly damaged.
C. Heart failure and cardiac arrest are different terms for the same condition, where the heart stops pumping blood effectively.
D. A primary symptom of congestive heart failure is the congestion of the lungs.

(1) A and B only
(2) B and C only
(3) A and D only
(4) A, C and D only

[Contd...

177. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Pulmonary ventilation (breathing) is driven by a pressure gradient established between the inside of the lungs and the external atmosphere.
Reason R: This gradient is generated by the contraction and relaxation of the diaphragm and the intercostal muscles, which alters thoracic volume.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

178. Which of the following statements about human platelets is **correct**?
(1) They are nucleated cells involved in oxygen transport.
(2) They are rich in clotting factors.
(3) They are nucleated cells that defend against pathogens.
(4) They are multinucleated cells that store heparin.

179. Which of the following is **not** a function of the conducting part?
(1) It clears the inhaled air of foreign particles.
(2) It brings the temperature of the inhaled air to body temperature.
(3) It provides the surface for the diffusion of O_2 and CO_2.
(4) It humidifies the atmospheric air before it reaches the lungs' exchange surface.

180. Given below are two statements.
Statement I: The outer pleural membrane is in close contact with the thoracic lining whereas the inner pleural membrane is in contact with the lung surface.
Statement II: The incomplete, C-shaped cartilaginous rings prevent the collapse of the respiratory passages during inspiration and allows for the expansion of the oesophagus during deglutition.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

■■■

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